

Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

A Lightweight Hybrid Machine Learning Framework for Phishing URL Detection and Classification

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Phishing attacks remain one of the most prevalent cybersecurity threats globally.

Traditional blacklisting methods are reactive and easily bypassed by zero-day attacks.

Complex deep learning models offer high accuracy but require significant computational overhead.

There is a critical need for a lightweight, robust, and highly accurate machine learning framework.

The proposed solution must generalize across different datasets and unknown environments.

Develop a hybrid ML framework combining Random Forest (RF) and Extreme Gradient Boosting (XGBoost).

Implement a Stacking Classifier with Logistic Regression as a meta-learner.

Extract lexical, structural, and NLP-based features directly from raw URLs without external requests.

Ensure low inference latency suitable for real-time edge or browser-level deployment.

Validate robustness through cross-dataset evaluation and 5-fold cross-validation.

Datasets & Preprocessing



PhiUSIIL Dataset: 134,850 URLs
with 54 features.

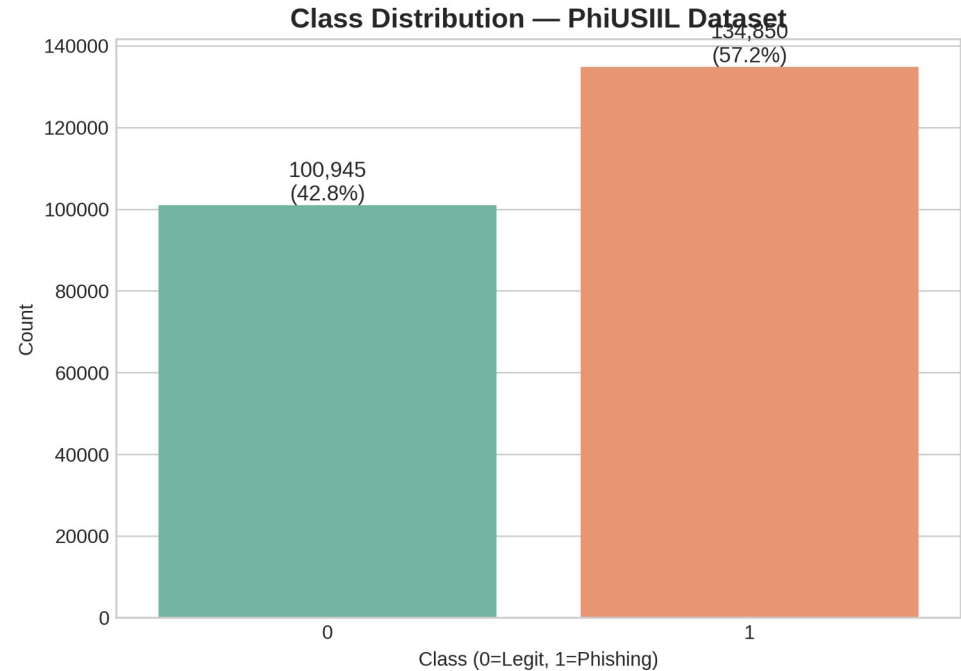
Web-Phishing Dataset: 73,575 URLs
with 39 features.

Data Integration: Consolidated
overlapping features into a
standardized 18-feature pipeline.

Label Mapping: 1 for Legitimate, 0
for Phishing.

Feature Engineering: Created TF-IDF
vectors for URL tokens.

Class Imbalance Handling: Utilized
Synthetic Minority Over-sampling



Tools and Technologies



Programming Language: Python
3.12+

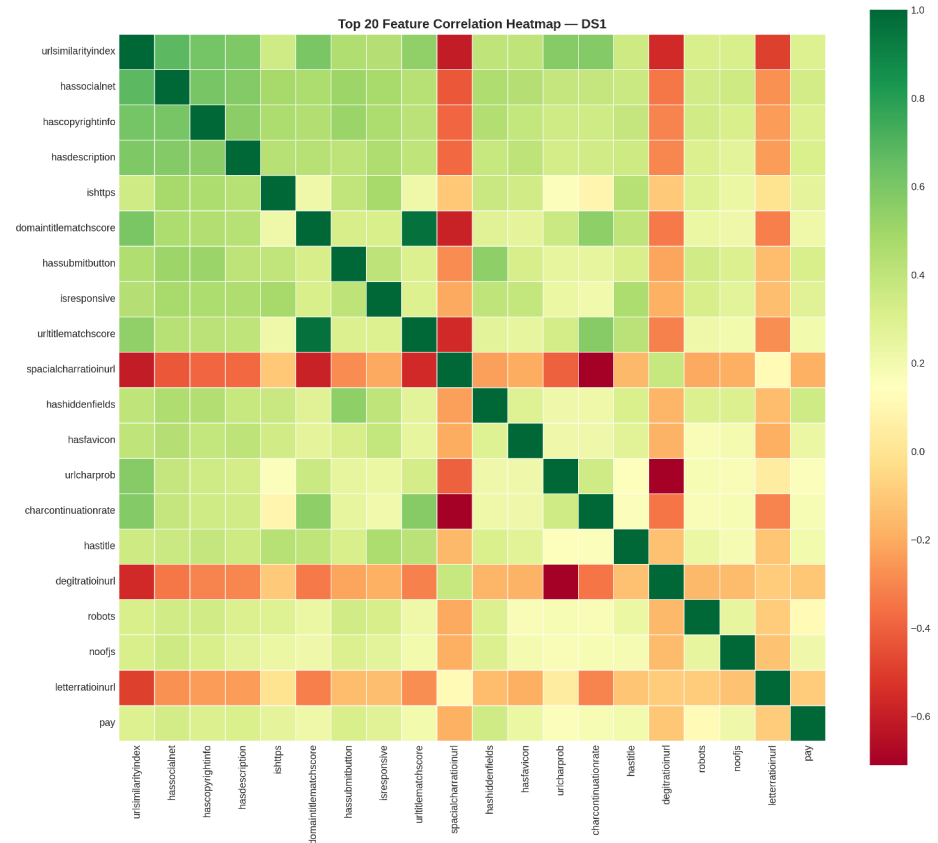
Data Processing: Pandas, NumPy

Machine Learning: Scikit-learn,
XGBoost

Model Interpretability: SHAP
(SHapley Additive exPlanations)

Visualization: Matplotlib, Seaborn

Environment & Ops: UV Package
Manager, Google Colab (for initial
compute)



1. URL Collection and Feature Extraction (Lexical & Length features).
2. Data Cleaning, Normalization, and SMOTE balancing.
3. Base Learner Training (RF, XGBoost) capturing diverse non-linear patterns.
4. Meta-Learner (Logistic Regression) aggregation for final classification.
5. Evaluation on local test splits and entirely unseen cross-datasets.

Performance Metrics (ROC)

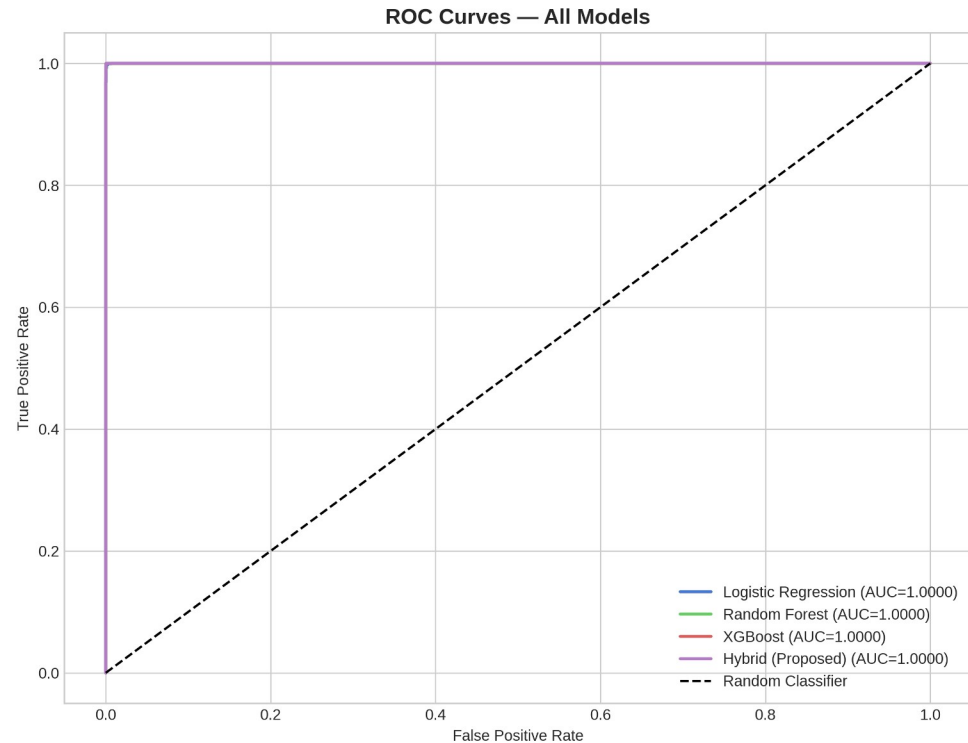


The Hybrid model achieved superior Area Under Curve (AUC).

Validation demonstrates the stacking approach effectively minimizes false positives.

Base learners (RF and XGB) provide strong individual predictions.

The ensemble approach mitigates the risk of overfitting.



Confusion Matrices

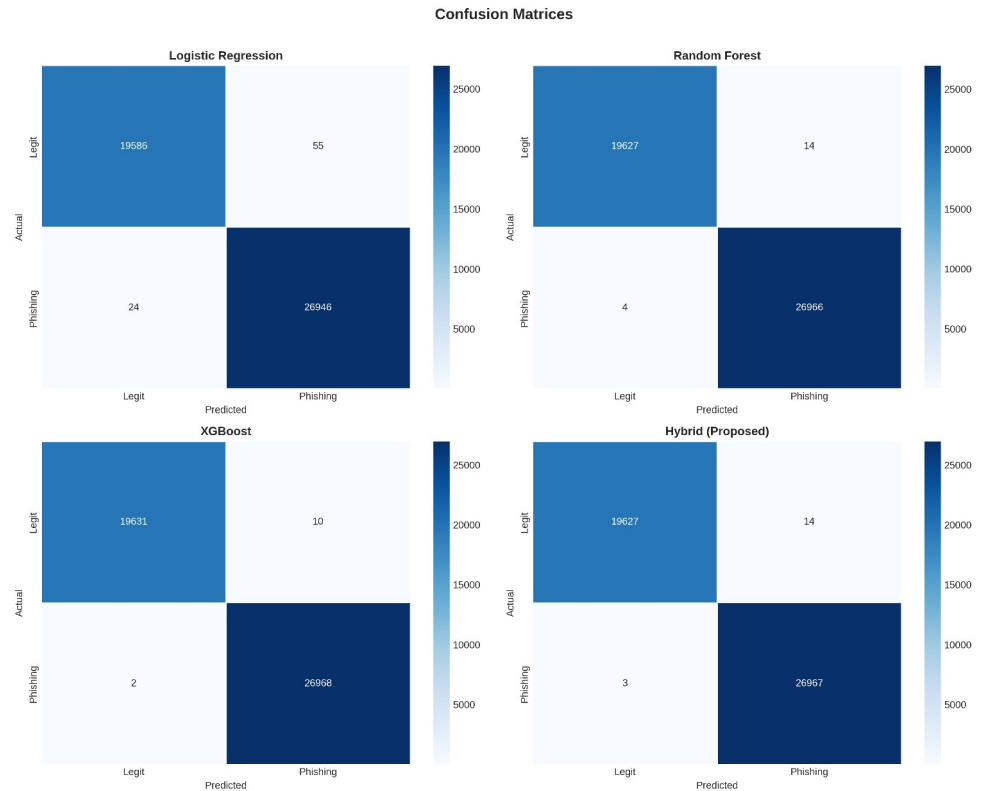


Excellent True Positive and True Negative rates.

Significantly lower False Positive Rate (FPR) compared to single algorithms.

Maintains high precision (98.9%) and F1-Score (98.7%).

Demonstrates balanced classification capability across both classes.



Feature Importance

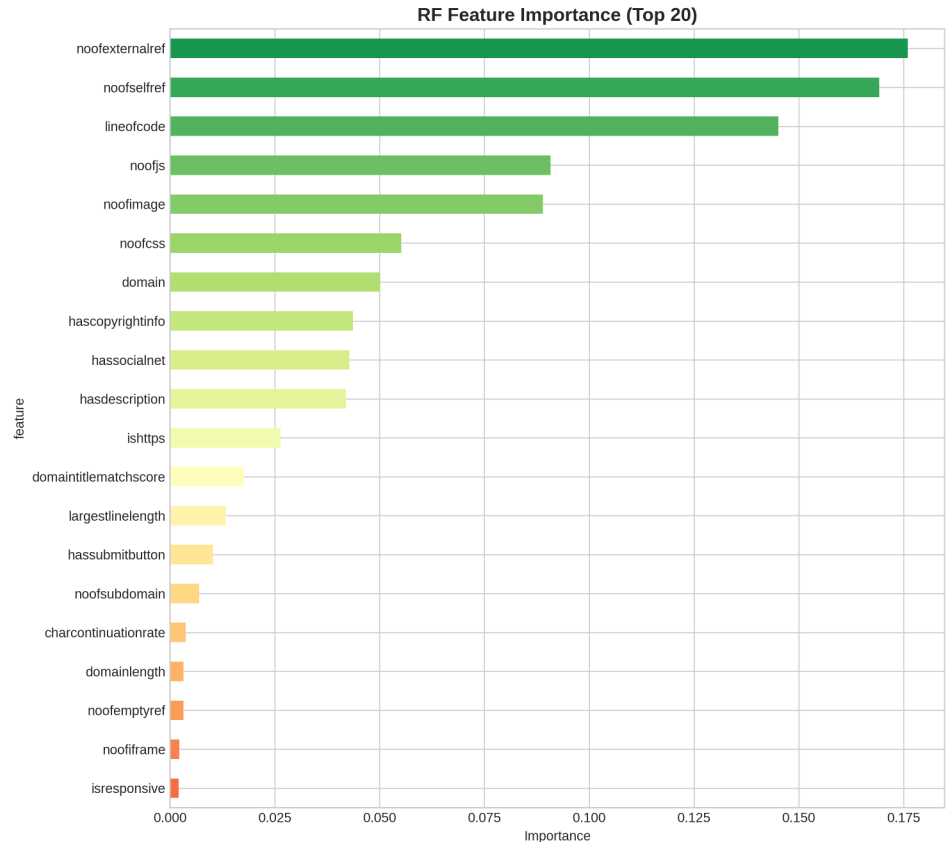


Identified top predictive features using Gini Importance.

Key Indicators: URL Length, Presence of HTTPS, Subdomain count.

Number of dots and special characters strongly correlate with phishing.

Helps in designing optimized, feature-reduced edge models.



Interpretability (SHAP)

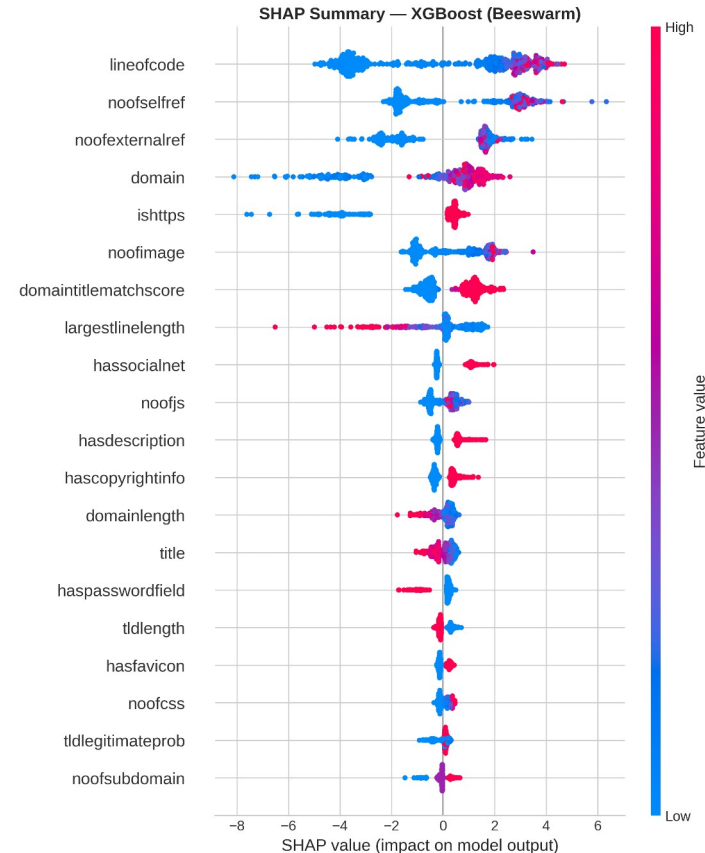


Used SHAP to ensure the model makes decisions logically (White-box AI).

SHAP Summary Plot shows impact of each feature on model output.

A long URL pushes the prediction towards Phishing (0).

Presence of secure flags pushes the prediction towards Legitimate (1).



Conclusion & Future Scope



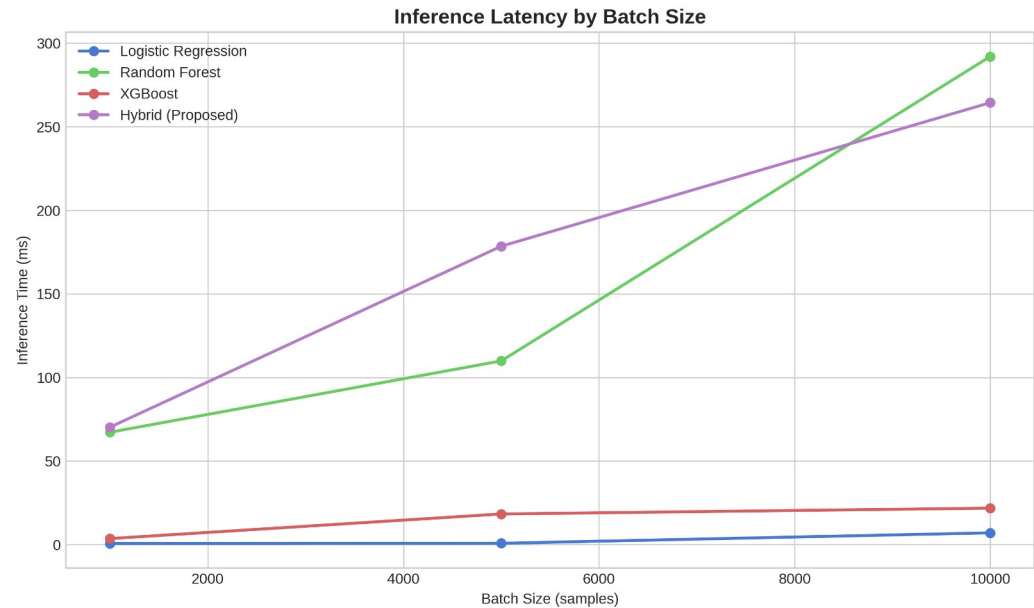
Conclusion: Developed a highly accurate (98.7%), lightweight Phishing Detector.

Cross-dataset testing proved high generalizability.

Future Scope: Real-time browser extension deployment.

Future Scope: API development for enterprise threat-intelligence integration.

Future Scope: Exploring deep learning for further URL NLP analysis.





Thank You

Any Questions?